

DATABASE SYSTEMS

Database Systems:

Over the past 50 years, a number of different approaches or models have represented accounting information systems. Each new model evolved because of the shortcomings and limitations of its predecessor. An interesting feature in this evolution is that the newest technique does not immediately replace older models. Thus, at any point in time, various generations of systems exist across different organizations and may even coexist within a single enterprise.

Database Systems

A modern auditor must understand how different Accounting Information System (AIS) approaches work. This knowledge helps the auditor examine and evaluate systems correctly. There are four main AIS models that an auditor may encounter:

- ❖ **Manual processes** – Accounting tasks are done by hand without using computers.
- ❖ **Flat-file systems** – Data is stored in separate files for each application.
- ❖ **Database approach** – Data is stored in a shared database that can be used by different applications.
- ❖ **Database Management System (DBMS)** – Specialized software is used to create, manage, and control access to databases

The Manual process model

The manual process model is the oldest and most traditional type of accounting system. In this model, business activities are performed manually and involve physical resources, documents, and people.

It includes activities such as taking customer orders, storing materials in warehouses, producing goods, delivering products to customers, and ordering goods from suppliers. Record-keeping in this system is also done by hand using paper documents.

Manual systems make it easier to understand internal control activities such as separation of duties, supervision, independent checking, audit trails, and control over access to records.

The Flat-File Model

The flat-file model is usually linked with legacy systems, which are old computer systems developed mainly between the late 1960s and the 1980 s. These systems were often run on large mainframe computers, and many organizations still use them today.

In a flat-file system, data is stored in separate files, and these files are not connected to each other. Each department or user keeps and controls their own data files instead of sharing data with others. As a result, the same data may be stored multiple times in different files.

Many old or legacy systems use the flat-file approach for managing data. In this system, each user or department owns and controls its own data files.

This exclusive ownership of data happens because of two main reasons:

➤ Business culture barriers

In the past, organizations often created strong boundaries between departments. These barriers prevented data from being shared across the entire organization.

➤ Technology limitations

Flat-file systems had technical limitations, so data files had to be designed to meet the specific needs of one main user. This made it difficult to share the same data with other users.

Figure: Flat-File Data Management

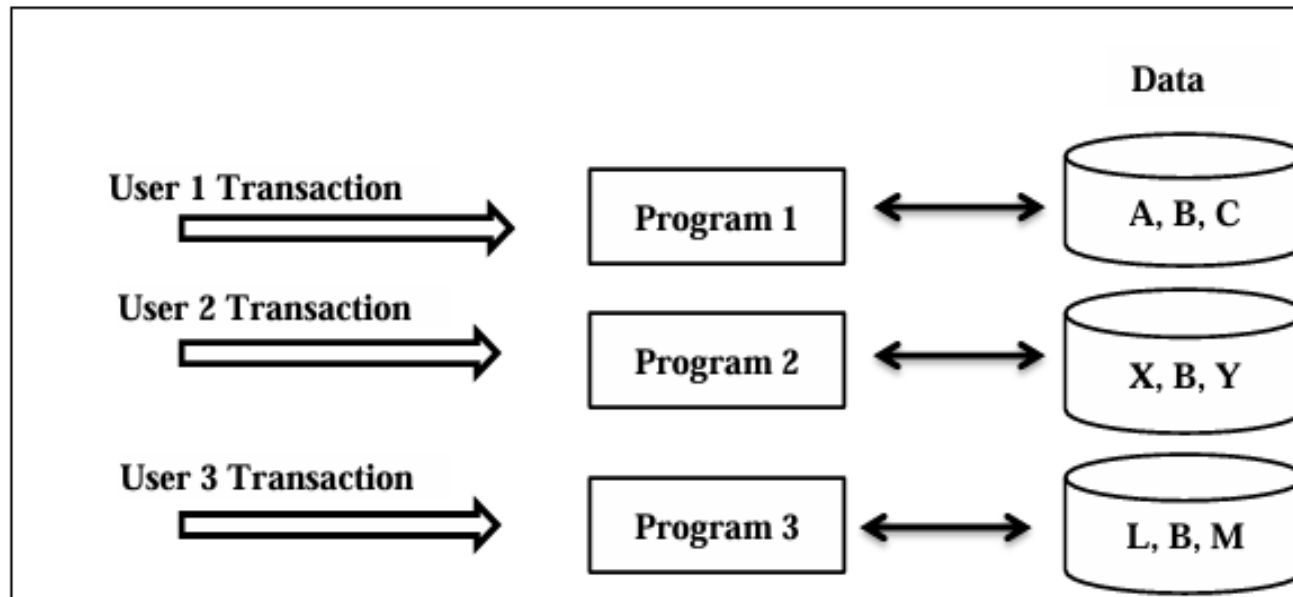


Figure 4.1., Flat-File Data Management

Flat-File Data Management

In the figure, the contents of files are shown using letters. Each letter may represent a data field, a record, or even a whole file. Notice that the same data element B appears in all user files. This situation is known as data redundancy, meaning the same data is stored in multiple places.

Data redundancy creates three major data management problems:

- 1. Data storage problems** – Extra storage space is needed because the same data is saved many times.
- 2. Data updating problems** – When data is changed, it must be updated in all files, which is time-consuming and error-prone.
- 3. Currency of information problems** – Some files may contain old or outdated data if updates are not done everywhere.

In addition to these, there is a fourth problem called task- data dependency, which is not directly caused by data redundancy. This problem will also be discussed separately.

The Database Approach

The biggest difference between the flat-file model and the database approach is that data is stored in one common database and shared by all users. Because everyone uses the same data, the problems of the flat-file system—such as data duplication and inconsistency—are greatly reduced. In the database approach, users do not own data individually. Instead, data sharing is the main idea, which helps ensure accuracy, consistency, and efficiency. The following are the main characteristics of a database:

- No data redundancy
- Single update
- Current values
- Task-data independence

The Database Management System

The main purpose of a Database Management System (DBMS) is to control and manage access to the database. A DBMS is special software that knows which data each user is allowed to see or use. When a user's program asks for data, the DBMS checks the user's permission level. If the user is authorized, the DBMS allows access to the data. If the user is not authorized, the DBMS blocks or denies access to that data.

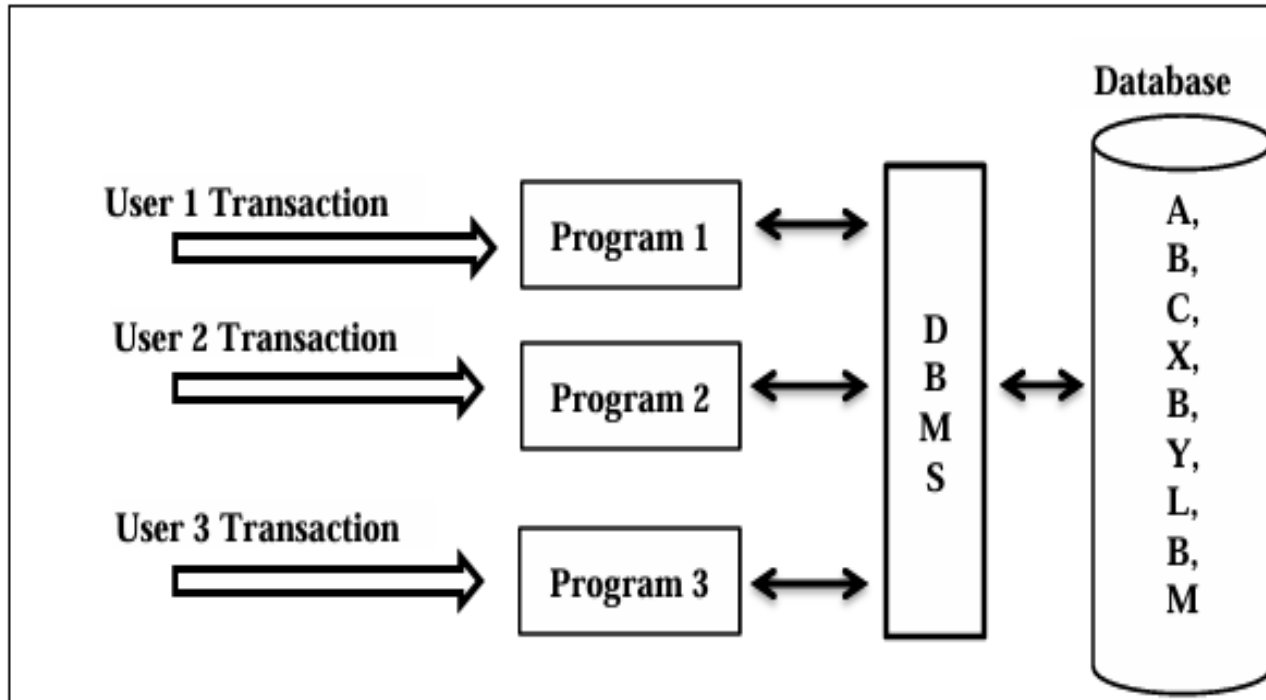


Figure 4.7., The Database Concept

Difference between File system & DBMS

File System	DBMS (Database Management System)
File system is a collection of files.	DBMS is a collection of related data.
User must write programs to manage data.	User does not need to write programs to manage data.
User must know how data is stored and represented.	DBMS hides storage and representation details.
Data storage and retrieval are inefficient.	Data storage and retrieval are efficient.
Has many problems with concurrent access.	Handles concurrent access using locking techniques.
Does not support crash recovery.	Supports crash recovery and data backup.
Data may be lost if system crashes.	Data is protected even after system failure.
Security and protection are weak.	Strong security and access control are provided.

Data Independence

Data independence means the ability to change the structure of data at one level of a database system without affecting the structure at the next higher level.

There are two types of data independence:

1. Logical Data Independence

Logical Data Independence is the ability to change the logical (conceptual) schema of a database without affecting the external schemas or application programs that access the data.

2. Physical Data Independence: Physical Data Independence is the ability to change the physical (internal) schema of a database without affecting the conceptual schema or application programs.

When not to use a DBMS

Using a Database Management System (DBMS) is not always the best choice. In some situations, its cost and complexity may outweigh its benefits.

Main disadvantages (costs) of using a DBMS

- High initial cost – A DBMS requires a large initial investment and may also need additional hardware.
- System overhead – Extra processing is needed to provide features such as security, data sharing, recovery, backup, and data integrity, which can reduce performance.

Situations where a DBMS may be unnecessary

- When the database and applications are simple, clearly defined, and unlikely to change.
- When there are strict real-time requirements that may not be met due to DBMS overhead.
- When multiple users do not need to access the data at the same time.
- When the DBMS cannot properly model very complex data.
- When users need special operations that the DBMS does not support.

Types of Databases and Database Applications

❑ Traditional Applications:

Numeric and Textual Databases

❑ More Recent Applications:

Multimedia Databases

Geographic Information Systems (GIS)

Data Warehouses

Real-time and Active Databases

Many other applications

Data Model

A model is a simplified representation that hides unnecessary details and shows only what is important. Data modeling is the process of representing important entities and the relationships between them in a database.

A data model is a set of concepts used to describe the structure of a database. It helps us understand how data is organized, stored, and related within the database. The structure of a database refers to the way data is held and arranged.

Types of Data Models

1. High Level- Conceptual data model.
2. Low Level – Physical data model.
3. Relational or Representational
4. Object-oriented Data Models
5. Object-Relational Models

- High Level-conceptual data model: User level data model is the high level or conceptual model. This provides concepts that are close to the way that many users perceive data.
- Low level-Physical data model : provides concepts that describe the details of how data is stored in the computer model. Low level data model is only for Computer specialists not for end-user.
- Representation data model: It is between High level & Low level data model Which provides concepts that may be understood by end-user but that are not too far removed from the way data is organized by within the computer.
- In Object Oriented Data Model, data and their relationships are contained in a single structure which is referred as object in this data model. In this, real world problems are represented as objects with different attributes.
- The Object-Relational data model refers to a combination of a Relational database model and an Object-Oriented database model. As a result, it supports classes, objects, inheritance, and other features found in Object-Oriented models, as well as data types, tabular structures, and other features found in Relational Data Models.

Schemas versus Instances

Schema and Instance in DBMS

In a Database Management System (DBMS), schema and instance are two important concepts used to organize and manage data. A schema is like the blueprint of the database, while an instance is the actual data stored in the database at a specific time.

What is Schema?

A schema is the overall design or structure of a database. It defines how data will be stored and organized. In DBMS, schema refers to the database architecture that shows how the database is constructed.

It describes:

- Tables and their structure

- Relationships between tables

- Rules and constraints on data

A schema acts as a template that determines how data is stored, arranged, and accessed in the database.

Example

Let's say a table teacher in our database named school, the teacher table requires the name, dob, and doj in their table so we design a structure as:

Teacher table

Name: String

Doj: date

Dob: date

Types of Schema

Schema is of three types: Logical Schema, Physical Schema and view Schema.

- **Logical Schema** – It describes the database designed at a logical level.
- **Physical Schema** – It describes the database designed at the physical level.
- **View Schema** – It defines the design of the database at the view level.

What is Instance?

An instance in a DBMS refers to the actual data stored in the database at a specific point in time . It represents the real values of the data that follow the structure defined by the schema.

While the schema defines the structure of the database, the instance shows the current contents of the database according to that structure. The instance changes frequently as data is added, updated, or deleted.

Example

Let's say a table teacher in our database whose name is School, suppose the table has 50 records so the instance of the database has 50 records for now and tomorrow we are going to add another fifty records so tomorrow the instance has a total of 100 records. This is called an instance.

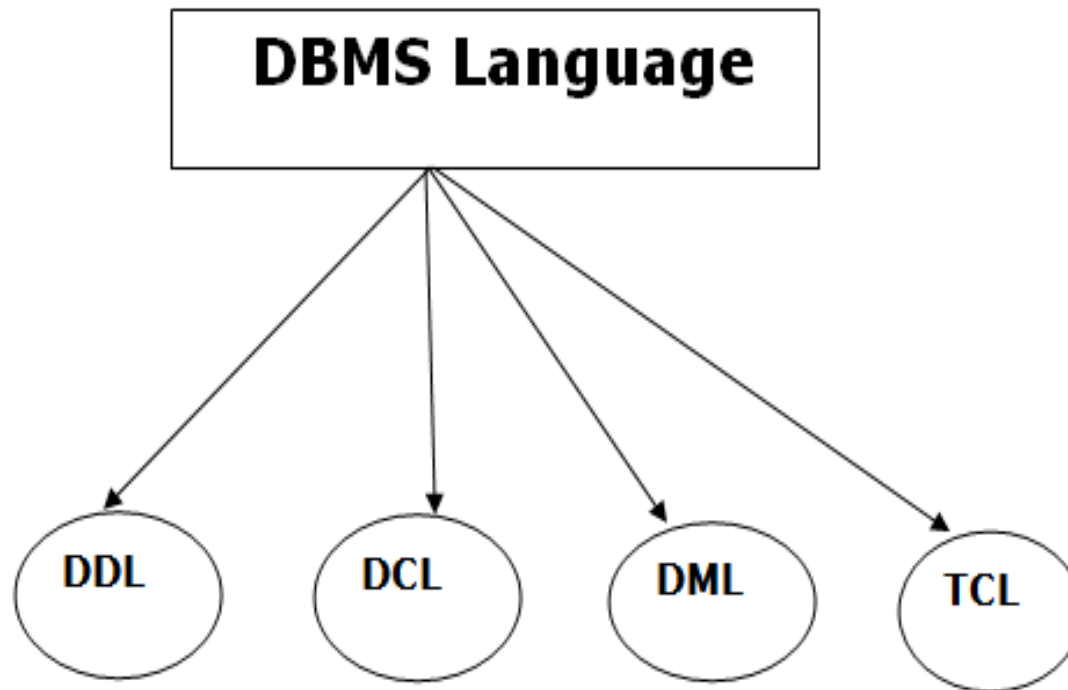
Difference Between Schema and Instance

Schema	Instance
It is the overall description of the database.	It is the collection of information stored in a database at a particular moment.
The schema is same for the whole database.	Data in instances can be changed using addition, deletion, and updating.
Does not change Frequently.	Changes Frequently.
Affects the entire database structure.	Affects only the current state of data.
Requires significant effort and planning to change.	Easily altered by performing CRUD (Create, Read, Update, Delete) operations.
Table structures, relationships, constraints.	Data entries, records in tables.

DBMS Languages

- A DBMS has appropriate languages and interfaces to express database queries and updates.
- Database languages can be used to read, store and update the data in the database.

Types of Database Languages



Data Definition Language (DDL)

- **DDL** stands for **Data Definition Language**. It is used to define database structure or pattern.
- It is used to create schema, tables, indexes, constraints, etc. in the database.
- Using the DDL statements, you can create the skeleton of the database.
- Data definition language is used to store the information of metadata like the number of tables and schemas, their names, indexes, columns in each table, constraints, etc.

Here are some tasks that come under DDL:

Create: Alter: Drop: Truncate: Rename: Comment:

Data Manipulation Language (DML)

- **DML** stands for **Data Manipulation Language**. It is used for accessing and manipulating data in a database. It handles user requests.

Here are some tasks that come under DML:

**Select: Insert: Update: Delete: Merge: Call:
Explain Plan: Lock Table:**

Data Control Language (DCL)

- DCL stands for Data Control Language. It is used to control access and permissions on data stored in a database, not to retrieve data. DCL helps the database administrator decide who can access what data.
- DCL commands are transactional, which means they support features like commit and rollback to manage changes safely.

Main DCL Commands

- GRANT :Used to give permission to users to access database objects (such as tables or views).
- REVOKE :Used to take back permissions that were previously given to users

Transaction Control Language (TCL)

- Transaction Control Language (TCL) is used to manage the changes made by Data Manipulation Language (DML) statements. It helps group multiple database operations into a logical transaction and ensures data consistency.

Main TCL Commands

- **COMMIT** : Saves all changes made in the current transaction permanently to the database.
- **ROLLBACK** : Undoes changes made in the current transaction and restores the database to its previous state.

Database Design Process

An organization's database is the main place where both financial and non-financial data are stored. The term database is used in a general sense—it can be a manual system like a filing cabinet or a computer-based system like a hard disk.

The database environment can be divided into four main elements:

1. **Users** – People who use the data and information.
2. **DBMS (Database Management System)** – Software that manages and controls access to the database.
3. **Database Administrator (DBA)** – The person responsible for managing, maintaining, and securing the database.
4. **Physical Database** – The actual place where data is stored.

Users

Users access the database in two ways.

✓ Through application programs

In this method, system professionals (programmers) create application programs. These programs send requests to the DBMS, which checks whether the request is valid and then retrieves the required data for processing.

✓ Through direct queries

In this method, users access the database directly without using any special application program. The DBMS provides a built-in query tool that allows authorized users to retrieve and process data on their own, without the help of programmers.

Database Management System

A Database Management System (DBMS) provides a controlled and secure environment for accessing and managing a database. It helps allow or restrict user access and ensures that data is handled efficiently.

A DBMS is software used to store, retrieve, and query data. It acts as an interface between users and the database, enabling users to create, read, update, and delete data safely and accurately.

Database Administrator

The Database Administrator (DBA) is responsible for managing and controlling the database. When many users share the same database, proper organization, coordination, rules, and guidelines are needed to keep the data accurate and secure. The DBA ensures the integrity and proper use of the database. In large organizations, the DBA role may be handled by an entire department of technical staff led by a database administrator. In smaller organizations, one person in the computer or IT services department may perform the DBA duties.. The duties of the DBA fall into the following areas:

- database planning,
- database design,
- database implementation,
- database operation and maintenance, and
- database change and growth.

The Physical Database

This is the lowest level of the database. The physical database consists of magnetic spots on magnetic disks. The other levels of the database (for example, the user view, conceptual view, and internal view) are abstract representations of the physical level. At the physical level, the database is a collection of records and files.

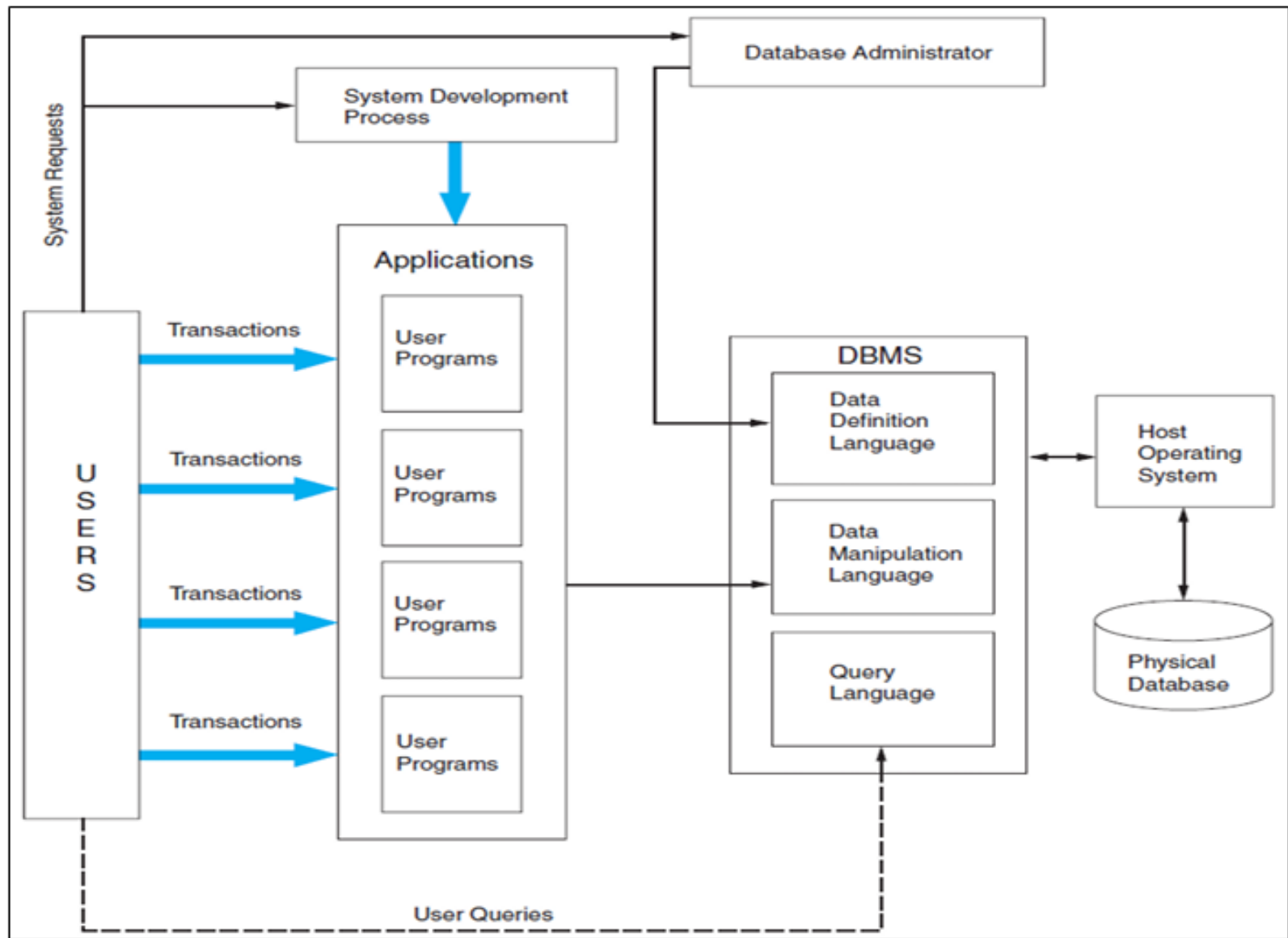


Figure 8.4: Elements of the Database Concept

Types of Database Model

- Hierarchical database model.
- Relational model.
- Network model.
- Object-oriented database model.
- Entity-relationship model.
- Document model.
- Entity-attribute-value model.
- Star schema.

The Relational (REA) Database Model

REA is an accounting framework that shows the important resources, events, and people (agents) in an organization and explains how they are connected to each other. After identifying these elements, both accounting data (like sales, cash) and non-accounting data (like customer details, employee information) can be collected and stored in a relational database. From this single shared database, different user views can be created to meet the needs of different users, such as managers, accountants, and auditors. Having multiple views makes transaction data more flexible and helps in building accounting information systems that are well integrated, rather than creating separate systems that act as barriers.

A system is relational if it:

- Represents data in the form of two-dimensional tables such as the database table, called Customer,
- Supports the relational algebra functions of restrict, project, and join.

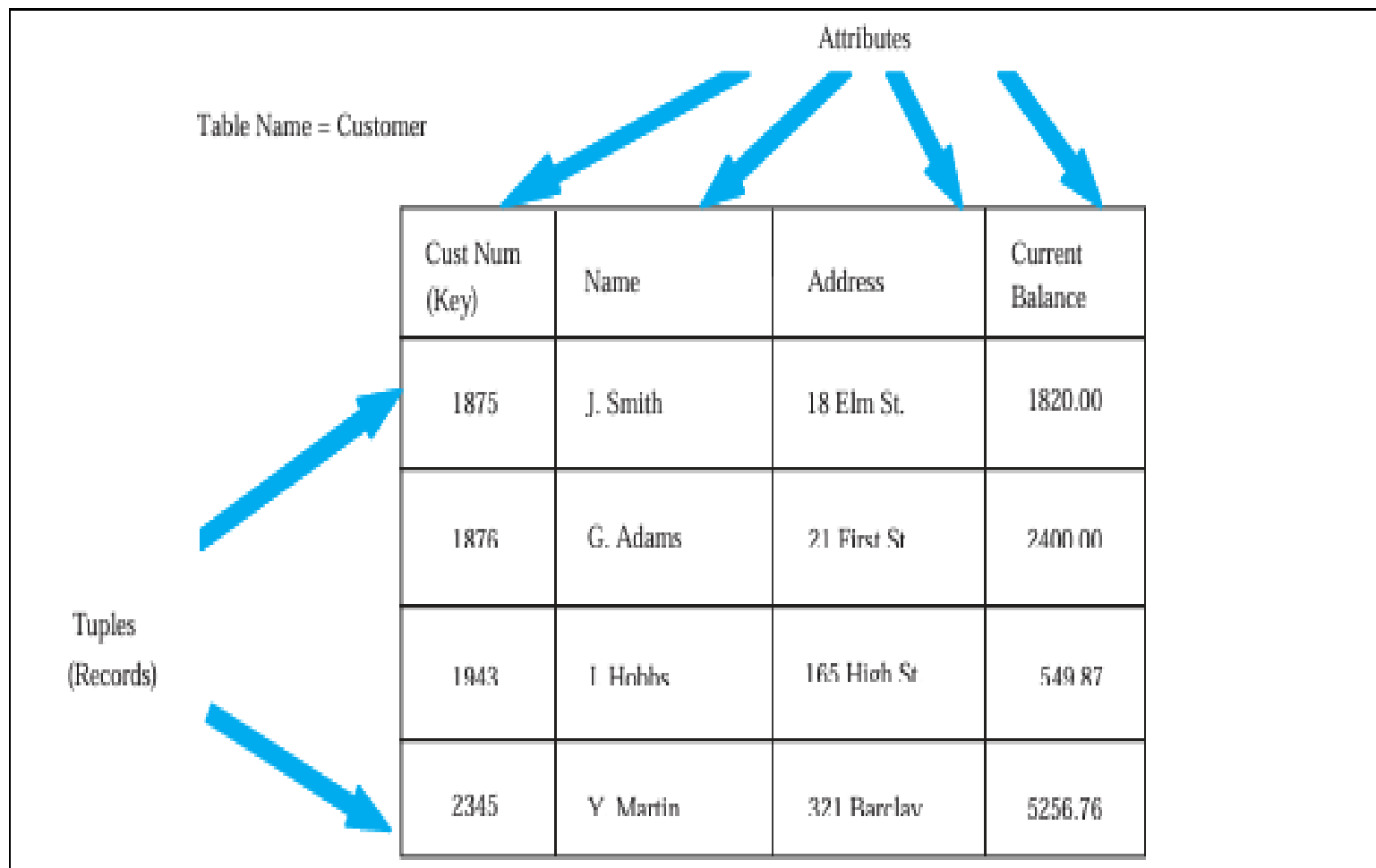
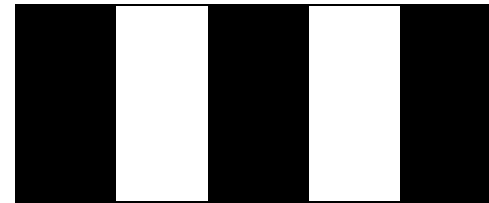


Figure 4.9: A Relational Table Called Customer

(a) Restrict



(b) Project



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(c) Join

Sales order	
Sales Order Number	Customer Name
1	X
2	Y
3	X

Customer balance	
Customer Name	Current balance
X	\$100
Y	\$200
Z	\$150

Sales order referring to customers balance		
Sales Order Number	Customer Name	Current balance
1	X	\$100
2	Y	\$200
3	X	\$100

Figure 4.10: The Relational Algebra Functions Restrict, Project, and Join

These three algebra functions are examined in the following section.

Restrict: Project: Join:

Although restrict, project, and join is not the complete set of relational functions, it is a useful subset that satisfies most business information needs.

The Elements of REA Database Model

